ANTS ANTS ANTS

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My implementation ended up being quite a bit different than what Jonathan and I had been discussing in class, because we ended up working alone after all. I went with a very probabilistic model that's not the most representative of real ants, but was a lot of fun to implement.

Ants "spawn" in an anthill, which is randomly placed on the screen. From there, they fan out, going in the cardinal directions (N, NE, E, etc). They explore semi-randomly until one of two things happens: they find the food (represented by the large green circle), or they die.

Spawning

There must be enough food in the nest for a new ant to be "born". The anthill begins with a stock of 20 food, enough to spawn 20 ants. As long as food remains in the anthill, there's a probability of spawning a new ant each frame. The probability goes down the less food there is, so an anthill with very little food may go a fairly long time without spawning a new ant.

Finding Food

If an ant finds food, they make a beeline (ant-line) to their anthill to drop off the food. At that point, the amount of food available in the anthill increases by five, meaning five more ants can spawn.

After the first ant has found food, it reports back the direction it was going when it found the food, represented by a probability vector (that was intended to be NESW but ended up being, instead, SWNE. oops). That direction is stored; new ants have an increasing probability of spawning going in the stored direction the more ants have returned with food.

If an ant returns having made the trip to the food and back in a shorter time than the previous ant that returned some food, the direction vector is updated to reflect that ant's direction they were going when they found the food. This means that over time, the ants will converge closer and closer to the optimal starting direction. However, as ant movement is semi-randomized, this will likely never quite be achieved, and could be thrown off even late in the run by random chance.

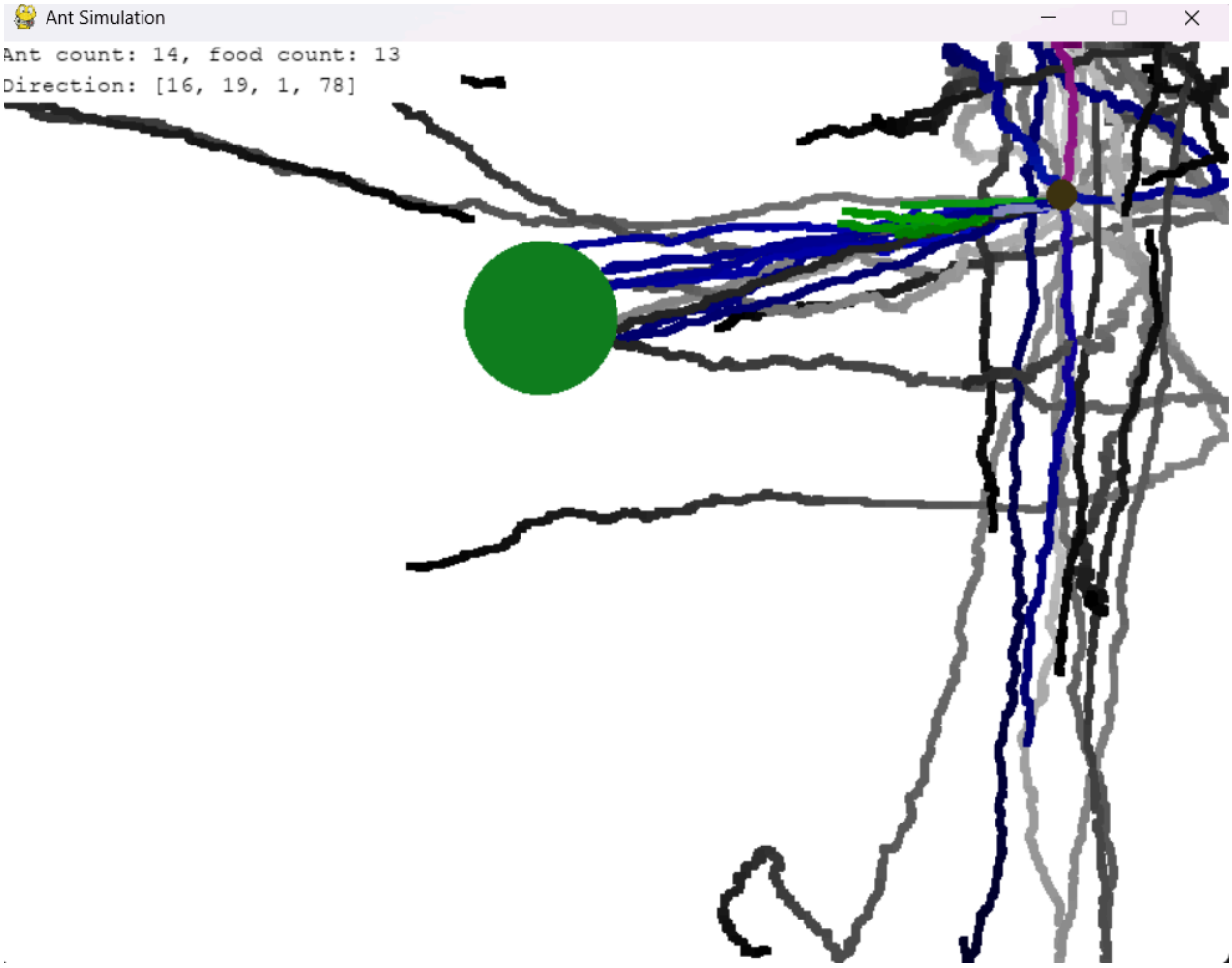
Food deliveries by an ant increase the anthill's food supply by 5, meaning 5 new ants can spawn. This number was chosen based on unscientific observation – running the simulation a few times, 4 or 5 seemed to be the sweet spot to where the ants have a probability of keeping the colony alive without being guaranteed a population explosion if they manage to bring any food back.

Lifespan

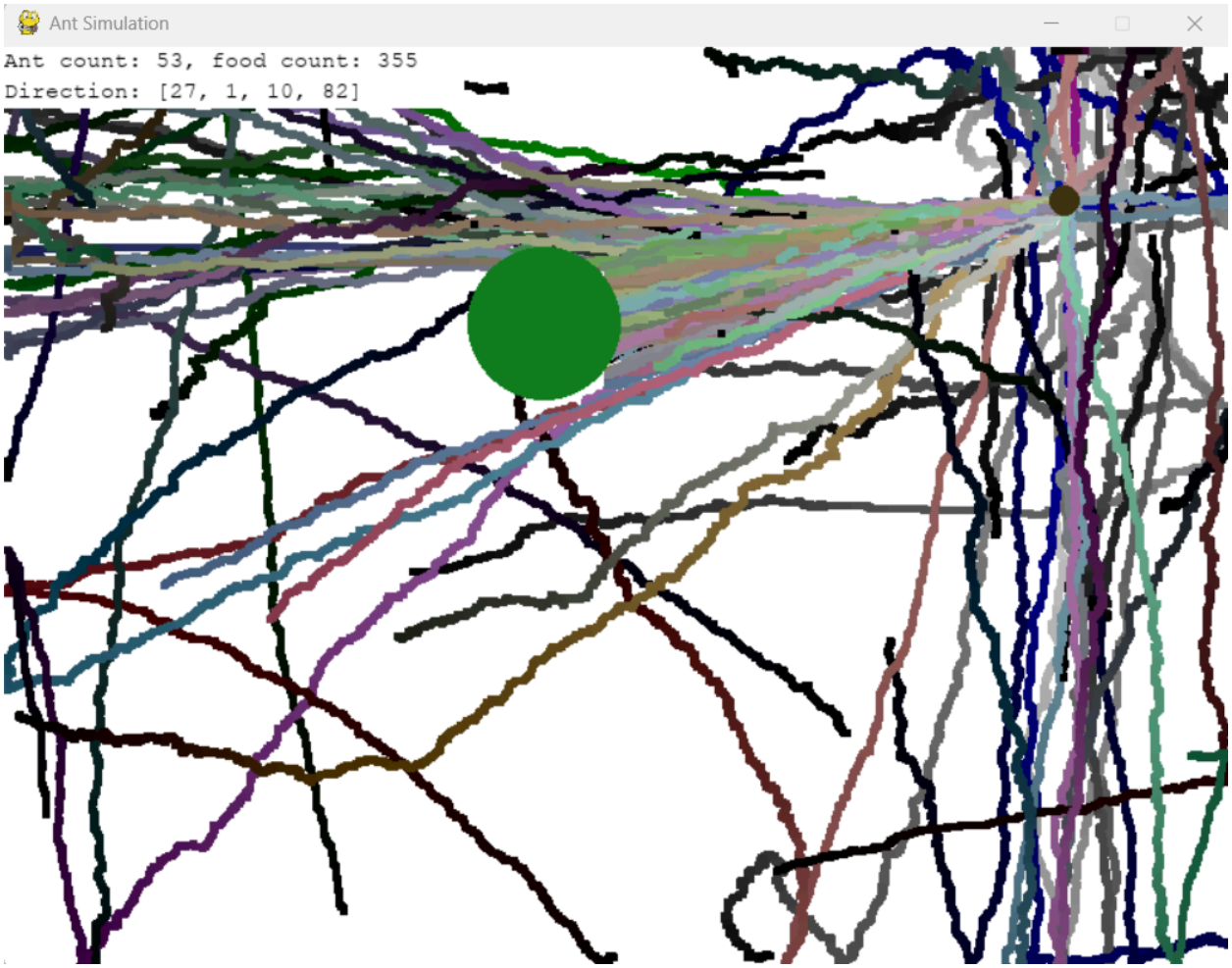
Each ant spawns with a different color, which represents their lifespan; they have a chance of getting darker with each frame, and when they turn fully black, they die. As an ant's lifespan decreases, its movement becomes more erratic, unless it finds the food; all ants can follow a nearly perfect path back to the anthill once they've found food.

Generation

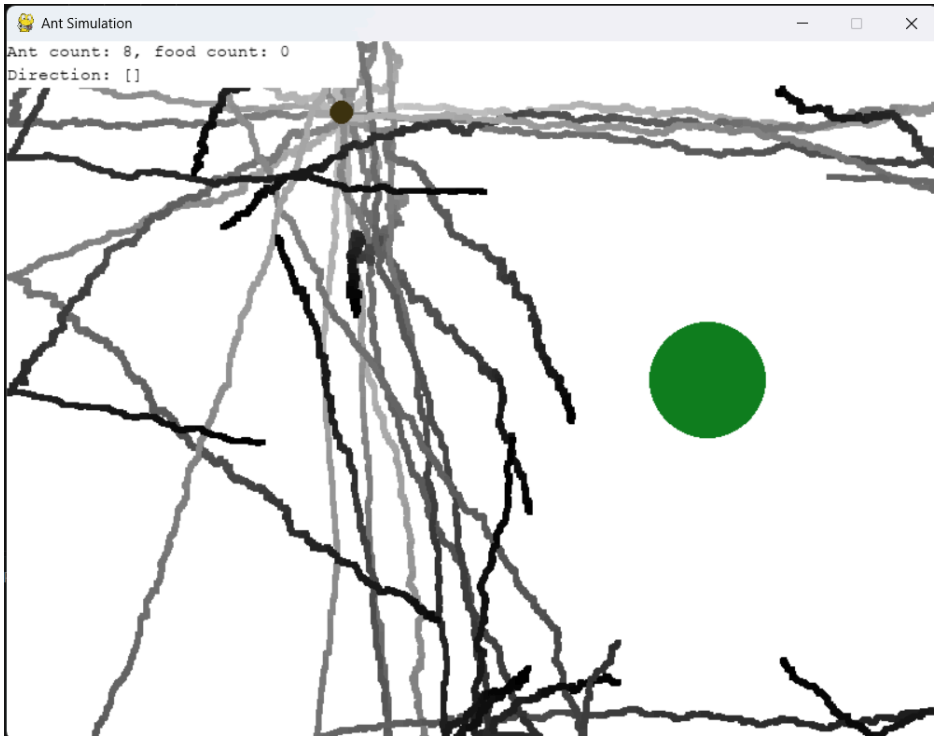
Mostly for my own ease of seeing whether the ants are behaving properly, I have each ant spawn with a semi-random starting color based on its "generation" – at what point it was born in the food-finding journey. Ants that are spawned before any food is found are all greyscale. Ants that spawn after the first piece of food is returned to the anthill are colored blue; if two pieces of food have been returned, the subsequent ants are colored purple; then green; once at least four pieces of food have been returned to the anthill, ants spawn with a random color.



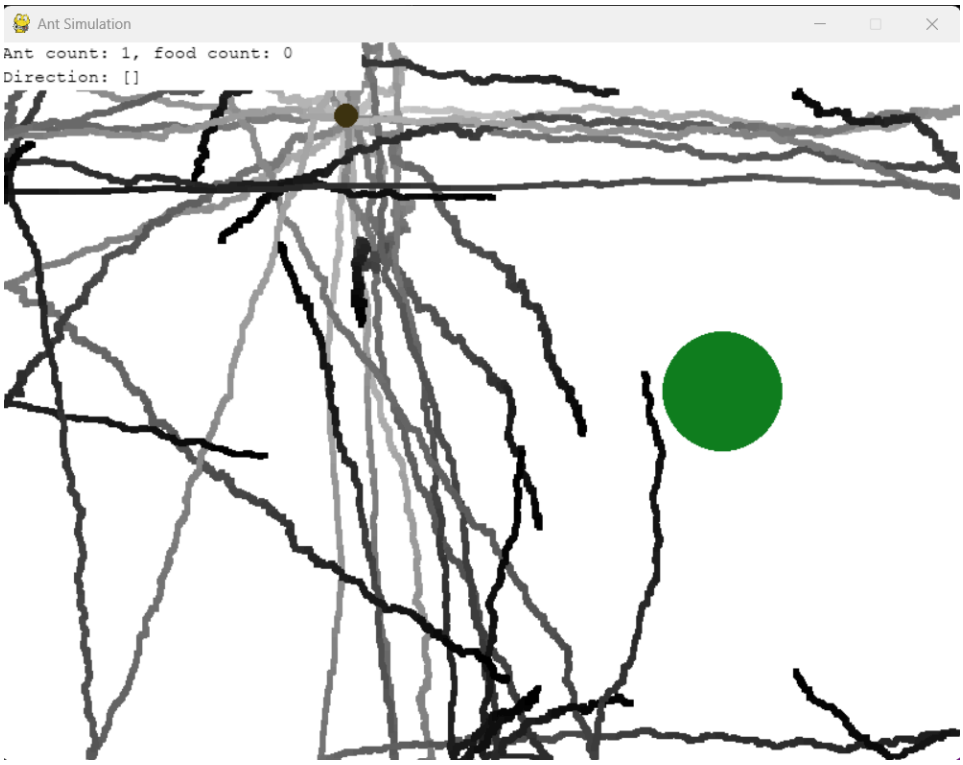
This screenshot shows an anthill that had a fairly easy time finding the food. They have a direction showing that they should go broadly west and a little bit south (and a little bit east, because ants are stupid and easily confused)



After a few round trips, the ants have found a better starting direction, which is overwhelmingly west and a little bit south. Most ant trails go more or less directly to the food and back, although some number of ants end up overshooting the nest, and some still go off in a random direction.



Sometimes the ants get unlucky and manage to miss the nest. In this simulation, all of the ants assigned to go southeast veered off course and ended up dying before they reached the food.



This colony ended up dying without ever finding the food, just due to random bad luck.